

The Transport Layer II (Part 2)

Detailed Flags, Structure, and Terminal Parsing

IT 213 Instructor

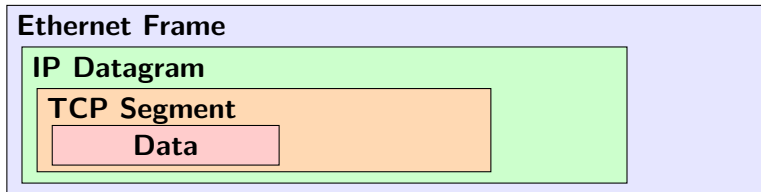
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Where does the TCP Header live?

The Concept: Encapsulation

- Your email (Data) is put inside a TCP box (Segment).
- The TCP box is put inside an IP box (Datagram).
- The IP box is put inside an Ethernet box (Frame).



The Control Panel: TCP Flags

Inside the header, there are 1-bit "switches" called **Flags**. Let's look at them in detail, grouped by their function.

Group 1: Connection Setup (The Handshake)

SYN (Synchronize)

- **Meaning:** "I want to start a new connection."
- **Used When:** Only in the very first packet.
- **Analogy:** Dialing a phone number.

ACK (Acknowledgment)

- **Meaning:** "I received your last message."
- **Used When:** In almost every packet after the start.
- **Analogy:** Saying "Uh-huh" or "Got it" while listening.

Detailed Flags: Data Flow

Group 2: Moving Data

PSH (Push)

- **Meaning:** "Don't wait! Send this to the application NOW."
- **Used When:** You are typing in a chat app or SSH. You don't want the computer to wait for a full packet to fill up; you want the letter 'A' sent immediately.

URG (Urgent)

- **Meaning:** "There is important data here (like a Cancel command)."
- **Status:** Rarely used in modern web traffic, but good to know it exists.

Detailed Flags: Closing & Emergency

Group 3: Ending the Call

FIN (Finish)

- **Meaning:** "I have no more data to send. I am hanging up."
- **Behavior:** This is a "polite" disconnect. Both sides wave goodbye.

RST (Reset) - The Emergency Brake

- **Meaning:** "ERROR! Stop talking to me!"
- **Used When:**
 - You try to connect to a port that is closed.
 - A Firewall blocks you.
 - The cable was unplugged and replugged suddenly.
- **Behavior:** Immediate disconnection. No goodbye.

How to "Parse" in the Terminal

Wireshark is great, but real admins often use the terminal. The tool we use is called **tcpdump**.

The Command: `sudo tcpdump -i any port 80`

Understanding the Output (The "Parsing"):

- tcpdump uses special symbols to show you the Flags.

Symbol	Flag	Meaning
[S]	SYN	Start Connection
[.]	ACK	Acknowledgment
[P]	PSH	Push Data
[F]	FIN	Finish/End
[R]	RST	Reset/Error

Example: Reading the Matrix

Let's parse a real line of output from tcpdump.

```
10:45:01 IP 192.168.1.5 > 8.8.8.8: Flags [S], seq 100
```

The Instructor's Translation:

- ❶ **IP ... > ...** : Source sending to Destination.
- ❷ **Flags [S]** : This is a **SYN** packet (Handshake Step 1).
- ❸ **seq 100** : My random starting Sequence Number is 100.

```
10:45:02 IP 8.8.8.8 > 192.168.1.5: Flags [S.], seq  
500, ack 101
```

Translation:

- ❶ **Flags [S.]** : This is **SYN** + **ACK** (Handshake Step 2).
- ❷ **ack 101** : "I received your #100, waiting for #101."

Deep Dive: The Sequence Number

We said TCP puts things "in order." But how?

Important Concept

TCP does NOT count "Packets" (1, 2, 3). **TCP counts BYTES.**

Example: Sending a picture (5000 bytes)

- **Packet 1:** Bytes 0-1000 → Seq: **0**
- **Packet 2:** Bytes 1001-2000 → Seq: **1001**

Summary: What to remember

① Flags Control Everything:

- **SYN/ACK:** Hello.
- **FIN:** Goodbye.
- **RST:** Error!

② Terminal Parsing:

- Look for [S], [F], [R] in tcpdump.

③ Sequence Numbers:

- We count bytes to ensure the file is perfect.

Next Class (Lab 5): We will use Wireshark to catch these [S] and [F] flags in real time!

The Final Piece: Ports

We have talked about IP Addresses (The Machine) and TCP (The Reliability). But how does the computer know which Application to give the data to?

The Problem: Your laptop is running Chrome, Spotify, and Discord all at the same time. When a packet arrives, who gets it?

The Solution: Ports (The Apartment Number)

- **IP Address:** The Street Address of the Building.
- **Port Number:** The Apartment Number inside.

"Deliver this packet to Building 192.168.1.5, Room 80."

The "Celebrity" Ports (Service Mapping)

Just like how "911" is always Emergency, certain Port numbers are reserved for specific famous services.

You must memorize these "Well-Known" Ports:

Port	Service	What it does
80	HTTP	Unencrypted Web (The old internet)
443	HTTPS	Secure Web (The Lock icon)
22	SSH	Secure Login (Linux Admin)
53	DNS	Domain Name System (UDP)
25	SMTP	Sending Email

If you try to connect to Port 80 using an Email client, the server won't understand you!

Server Ports vs. Client Ports

There is one tricky detail.

① Server Ports (Fixed):

- The Web Server **MUST** be on Port 80.
- Why? So everyone knows where to knock.

② Client Ports (Random/Ephemeral):

- When you open Chrome, your computer picks a **random** port (e.g., 54321) as the "Source Port."
- Why? So when Google replies, it knows exactly which browser tab to send the data back to.

The Conversation:

Source: 192.168.1.5:54321 → Dest: 8.8.8.8:80
"Hi Google (80), this is Chrome (54321). Send me the homepage."